

SEOUL AB, Korea, Republic of

WMO: 471110

Lat: 37.446N

Lon: 127.114E

Elev: 28

StdP: 100.99

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.5	-10.2	-21.9	0.5	-8.6	-20.0	0.6	-7.1	6.1	-0.7	5.4	-1.6	2.0	330	0.400

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.8	33.8	25.8	32.1	24.9	30.9	24.2	26.7	31.1	26.2	30.0	25.6	29.1	2.7	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.9	21.3	28.6	25.1	20.3	28.1	24.3	19.3	27.5	84.2	31.6	81.5	29.8	78.8	29.0	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.1	5.3	4.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.7	-2.3	0.2	6.1	12.1	17.8	22.7	25.9	26.5	21.2	14.4	7.7	-0.4
	DBStd	10.53	4.04	4.40	3.58	3.71	2.78	2.34	2.43	2.51	2.84	3.32	4.15	4.51
	HDD10.0	1227	382	274	129	21	0	0	0	0	0	6	91	324
	HDD18.3	2796	640	507	378	188	43	2	0	0	8	128	320	582
	CDD10.0	2221	0	1	10	84	243	380	492	513	335	142	21	0
	CDD18.3	747	0	0	0	2	28	132	233	254	93	5	0	0
	CDH23.3	6341	0	0	1	28	277	955	2041	2431	578	31	0	0
	CDH26.7	2116	0	0	0	2	51	271	721	951	119	2	0	0
(14) Wind	WSAvg	1.9	1.7	1.9	2.3	2.3	2.1	1.8	2.0	1.9	1.5	1.6	1.7	1.8
(15) Precipitation	PrecAvg	1372	19	29	44	82	94	144	387	289	168	48	44	25
	PrecMax	2138	63	78	130	189	240	432	877	880	586	185	128	58
	PrecMin	802	1	1	4	20	12	29	126	61	20	1	7	4
	PrecStd	312	15	21	29	42	45	93	174	190	147	39	26	15
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	14.9	20.4	26.2	30.1	32.8	35.2	35.8	31.0	26.2	20.2	12.2
		MCWB	5.0	11.6	12.2	15.3	18.8	21.8	25.3	26.3	23.4	19.6	14.0	8.1
	2%	DB	7.1	11.9	17.1	24.0	28.0	30.9	33.2	34.1	29.2	23.9	17.9	10.0
		MCWB	3.2	7.1	10.1	15.2	17.6	21.4	25.5	26.4	22.1	17.6	12.9	6.7
	5%	DB	5.8	9.2	14.9	22.0	26.2	29.6	31.9	32.9	28.0	22.1	16.1	8.1
		MCWB	2.3	5.2	8.6	13.1	16.9	20.8	25.0	25.8	21.4	16.2	11.6	5.4
	10%	DB	4.0	7.1	12.9	19.9	24.8	28.1	30.2	31.2	26.8	20.8	14.2	6.8
		MCWB	1.1	3.4	7.3	12.1	16.4	20.4	24.5	25.2	20.8	15.5	10.6	4.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.1	12.9	13.9	18.3	21.5	24.2	27.3	27.9	25.6	20.8	16.1	9.6
		MCDB	7.7	13.8	17.7	23.2	26.7	28.8	32.1	32.5	28.8	23.9	18.7	10.7
	2%	WB	4.0	7.7	11.6	16.0	19.5	23.2	26.5	27.1	24.2	19.2	13.9	7.4
		MCDB	6.1	10.1	15.7	22.1	24.7	27.8	30.7	31.7	27.1	22.1	16.7	9.1
	5%	WB	2.6	5.5	9.6	14.4	18.5	22.5	25.9	26.5	23.2	17.6	12.6	5.6
		MCDB	5.0	8.5	13.8	19.8	23.7	26.9	29.6	30.7	26.1	20.5	15.2	7.7
	10%	WB	1.3	3.9	7.7	13.1	17.4	21.7	25.4	26.0	22.2	16.2	11.2	4.1
		MCDB	3.7	6.8	11.8	18.2	22.8	25.9	28.9	29.7	25.2	19.6	13.8	6.0
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	9.4	10.5	11.7	11.6	9.9	7.5	7.8	9.6	11.0	9.5	8.6
		MCDBR	10.3	12.0	13.8	15.0	14.5	12.2	9.8	9.5	10.5	11.9	11.4	9.8
		MCWBR	7.1	7.7	7.8	7.5	6.6	4.9	3.9	3.7	4.5	6.3	7.2	7.3
	5% WB	MCDBR	9.1	10.2	11.9	12.3	11.2	9.4	7.6	8.0	8.1	9.8	9.9	8.9
		MCWBR	7.1	7.6	7.7	7.1	6.3	4.4	3.5	3.6	4.1	5.9	7.3	7.4
(40) Clear-Sky Solar Irradiance	taub		0.444	0.498	0.584	0.628	0.640	0.712	0.678	0.591	0.516	0.508	0.489	0.440
	taud		1.931	1.837	1.694	1.647	1.669	1.592	1.708	1.905	2.034	1.937	1.927	1.962
	Ebn at Noon		688	702	683	685	688	639	657	706	737	687	636	657
	Edh at Noon		141	176	223	247	246	265	235	189	157	157	142	129
(44) All-Sky Solar Radiation	RadAvg		2.49	3.26	4.24	4.95	5.57	5.20	4.05	4.30	4.10	3.57	2.44	2.19
	RadStd		0.15	0.26	0.43	0.36	0.51	0.52	0.61	0.61	0.46	0.26	0.24	0.15

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	+50

Nomenclature: See separate page